Generative Ai And Ai Platform Engineering

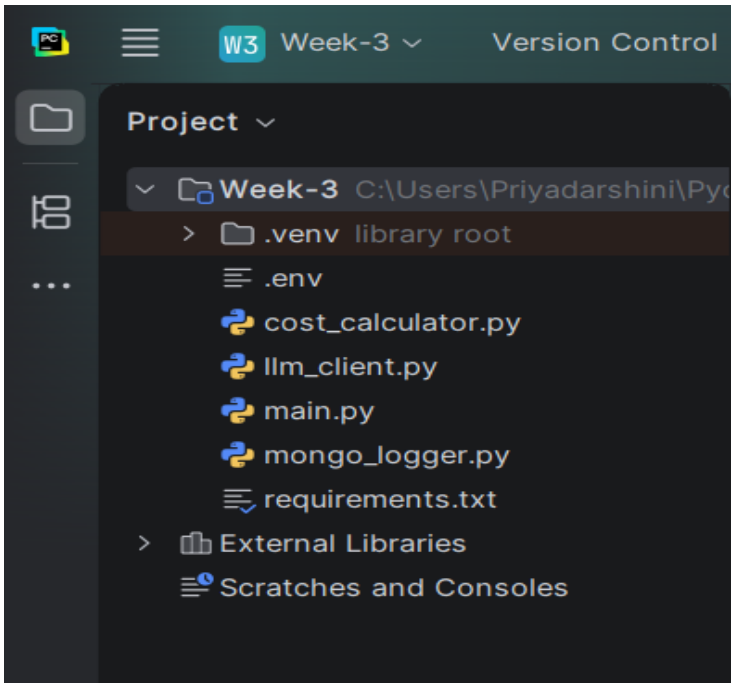
Name:- K. Priyadarshini

Roll no:- 2520030067

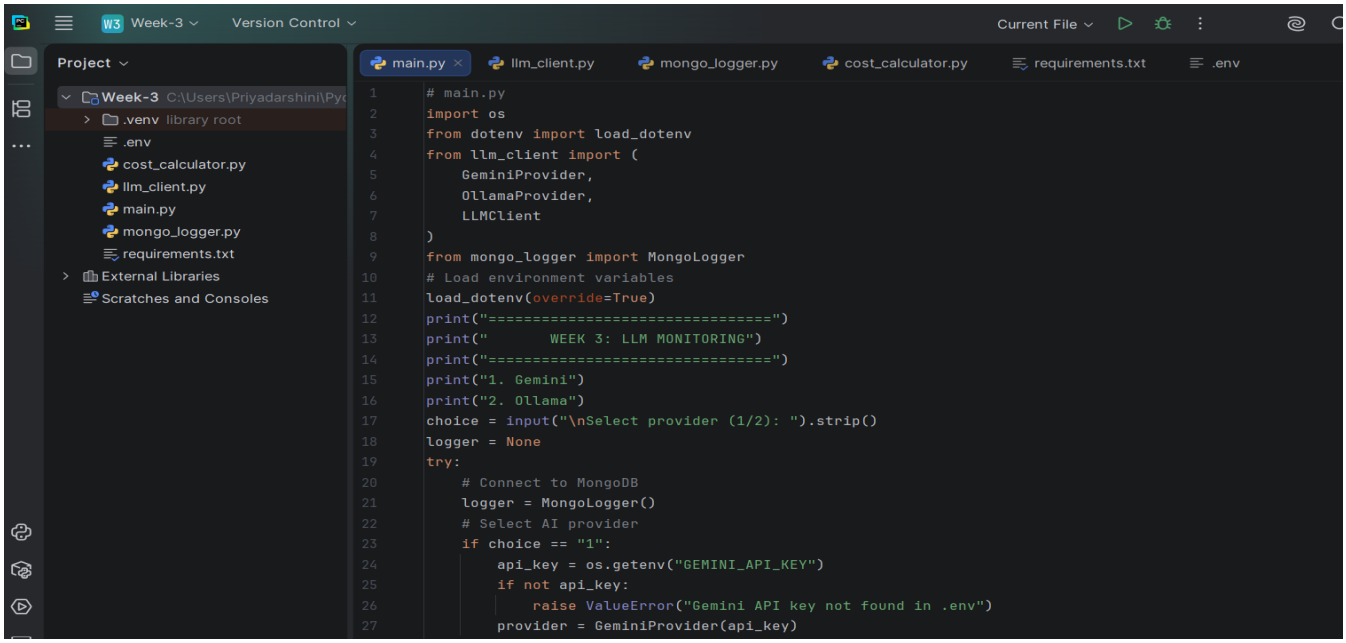
Section no:- 3

WEEK-1

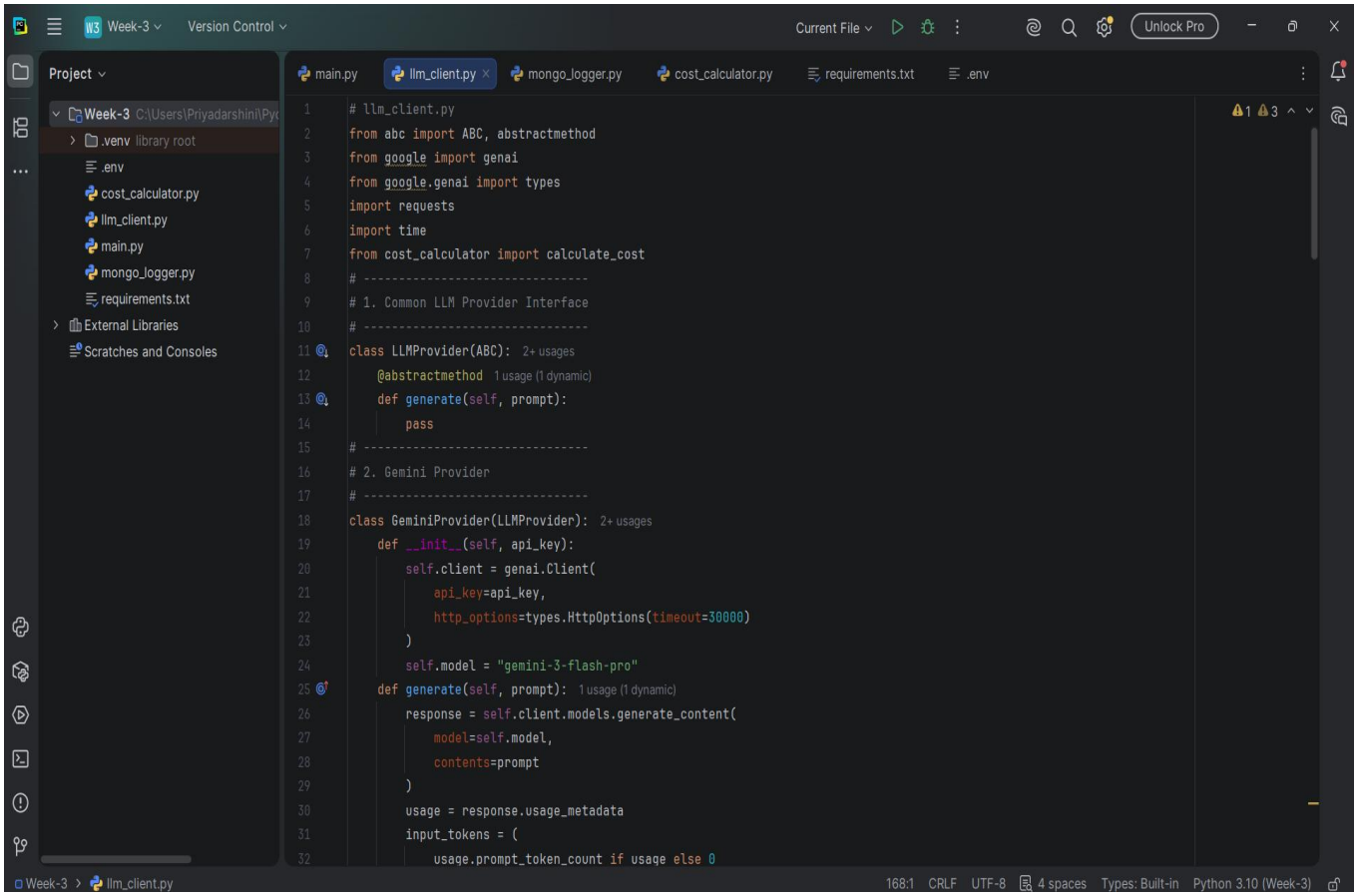
Step 1:-



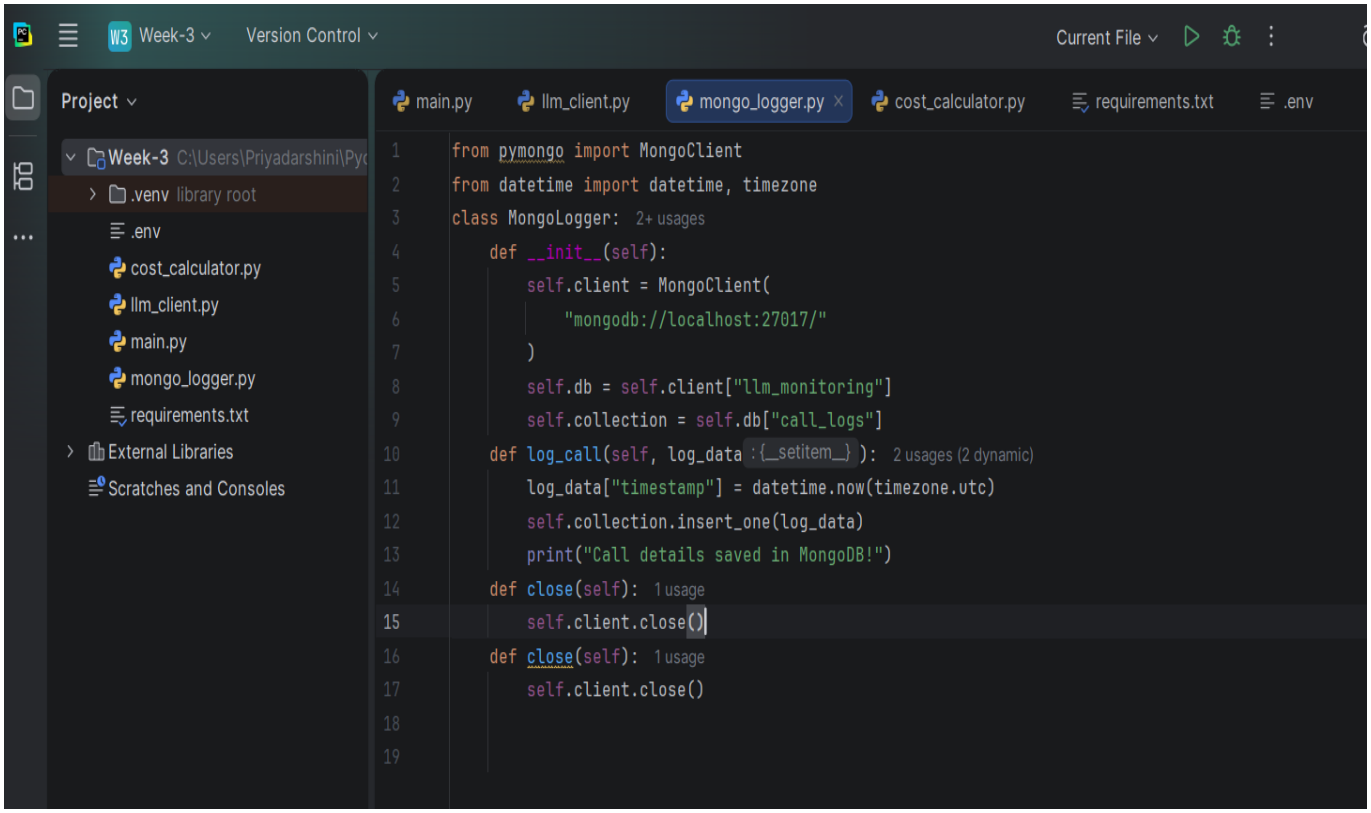
Step 2:-



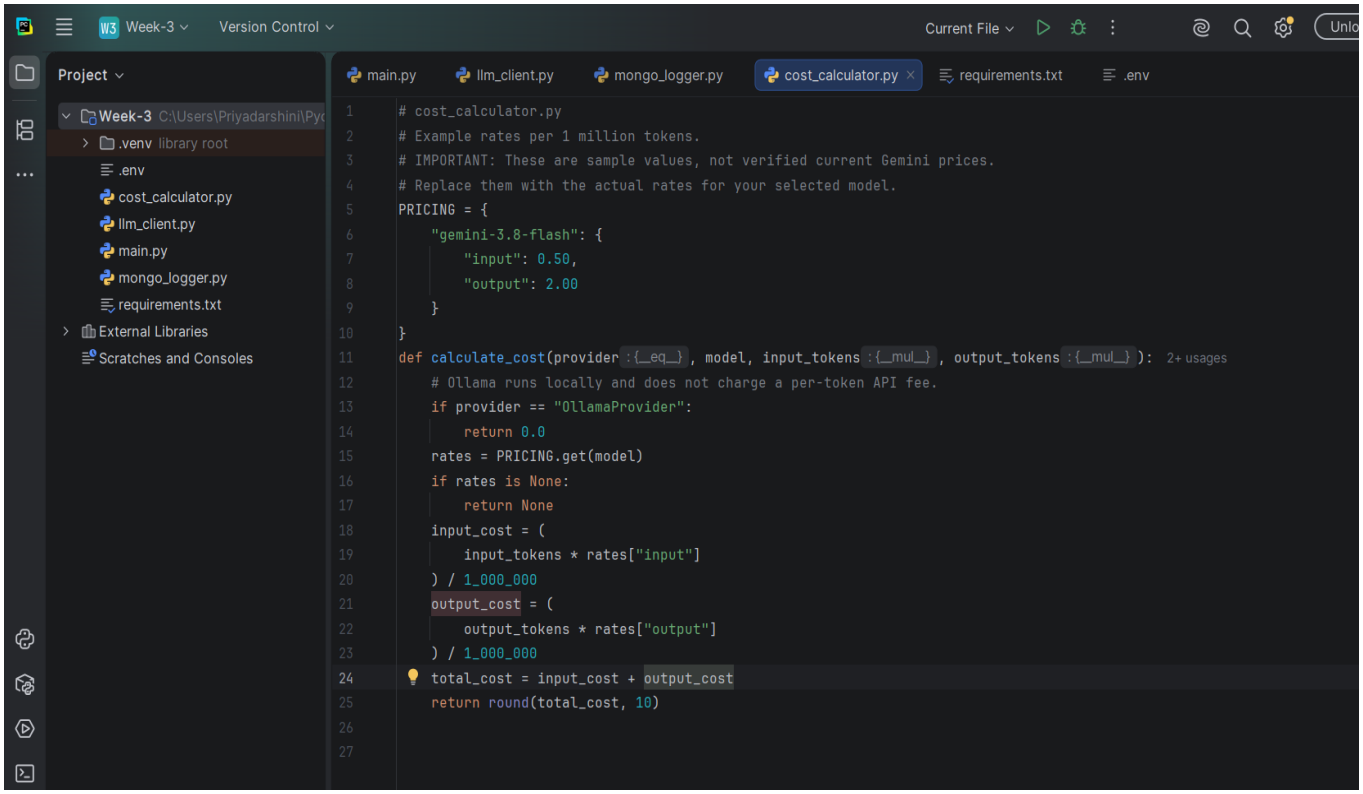
Step 3:-


Week-3
Version Control
Current File
Unlock Pro
Project
main.py
llm_client.py
mongo_logger.py
cost_calculator.py
requirements.txt
.env
Week-3 C:\Users\Priyadarshini\Py
.venv library root
.env
cost_calculator.py
llm_client.py
main.py
mongo_logger.py
requirements.txt
External Libraries
Scratches and Consoles
llm_client.py
from abc import ABC, abstractmethod
from google import genai
from google.genai import types
import requests
import time
from cost_calculator import calculate_cost
---
1. Common LLM Provider Interface
---
class LLMProvider(ABC): 2+ usages
@abstractmethod 1 usage (1 dynamic)
def generate(self, prompt):
pass
---
2. Gemini Provider
---
class GeminiProvider(LLMProvider): 2+ usages
def __init__(self, api_key):
self.client = genai.Client(
api_key=api_key,
http_options=types.HttpOptions(timeout=30000)
)
self.model = "gemini-3-flash-pro"
def generate(self, prompt): 1 usage (1 dynamic)
response = self.client.models.generate_content(
model=self.model,
contents=prompt
)
usage = response.usage_metadata
input_tokens = (
usage.prompt_token_count if usage else 0
Week-3 > llm_client.py
168:1 CRLF UTF-8 4 spaces Types: Built-in Python 3.10 (Week-3)

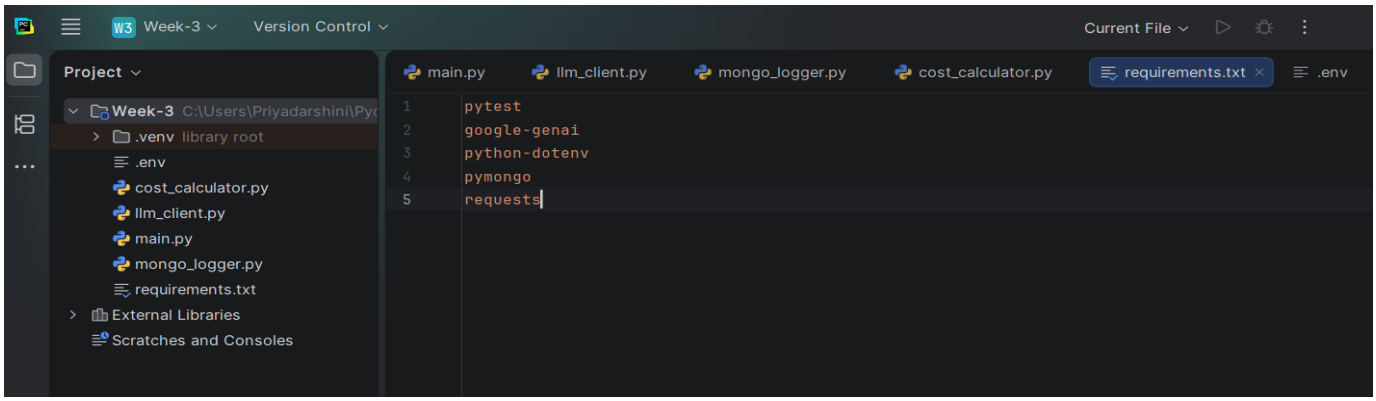
Step 4:


Week-3
Version Control
Current File
Project
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requirements.txt
.env
Week-3 C:\Users\Priyadarshini\Py
.venv library root
.env
cost_calculator.py
llm_client.py
main.py
mongo_logger.py
requirements.txt
External Libraries
Scratches and Consoles
from pymongo import MongoClient
from datetime import datetime, timezone
class MongoLogger: 2+ usages
def __init__(self):
self.client = MongoClient(
"mongodb://localhost:27017/"
)
self.db = self.client["llm_monitoring"]
self.collection = self.db["call_logs"]
def log_call(self, log_data : {__setitem__}): 2 usages (2 dynamic)
log_data["timestamp"] = datetime.now(timezone.utc)
self.collection.insert_one(log_data)
print("Call details saved in MongoDB!")
def close(self): 1 usage
self.client.close()
def close(self): 1 usage
self.client.close()

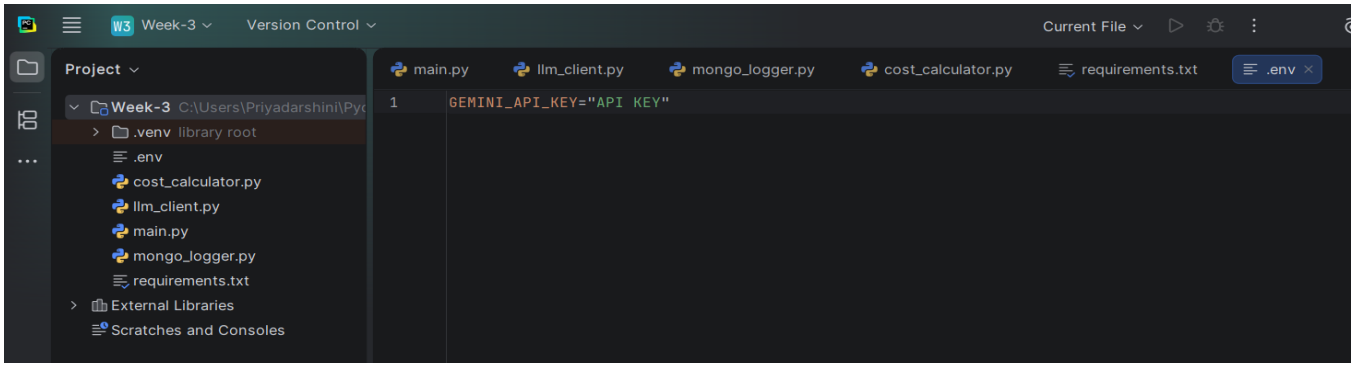
Step 5:-



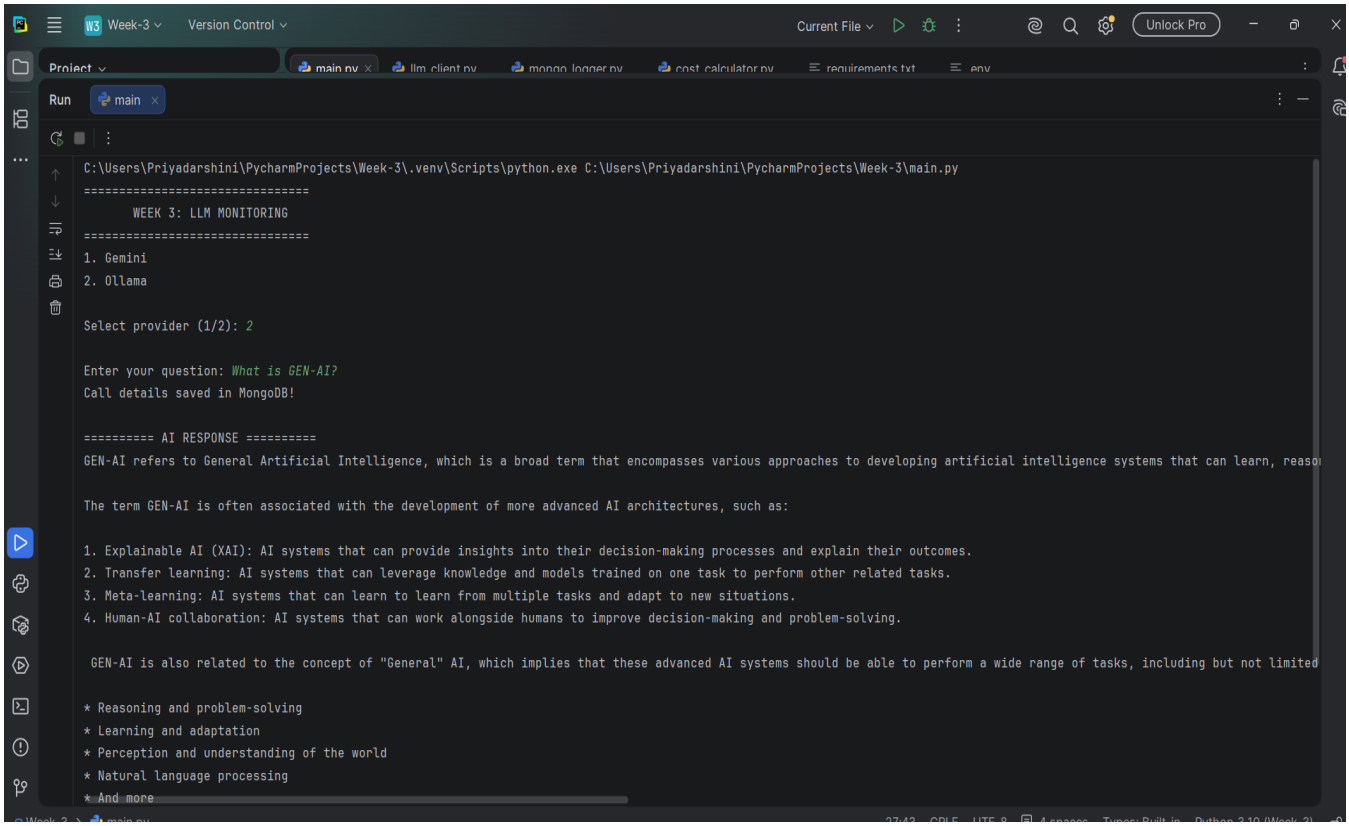
Step 6:-

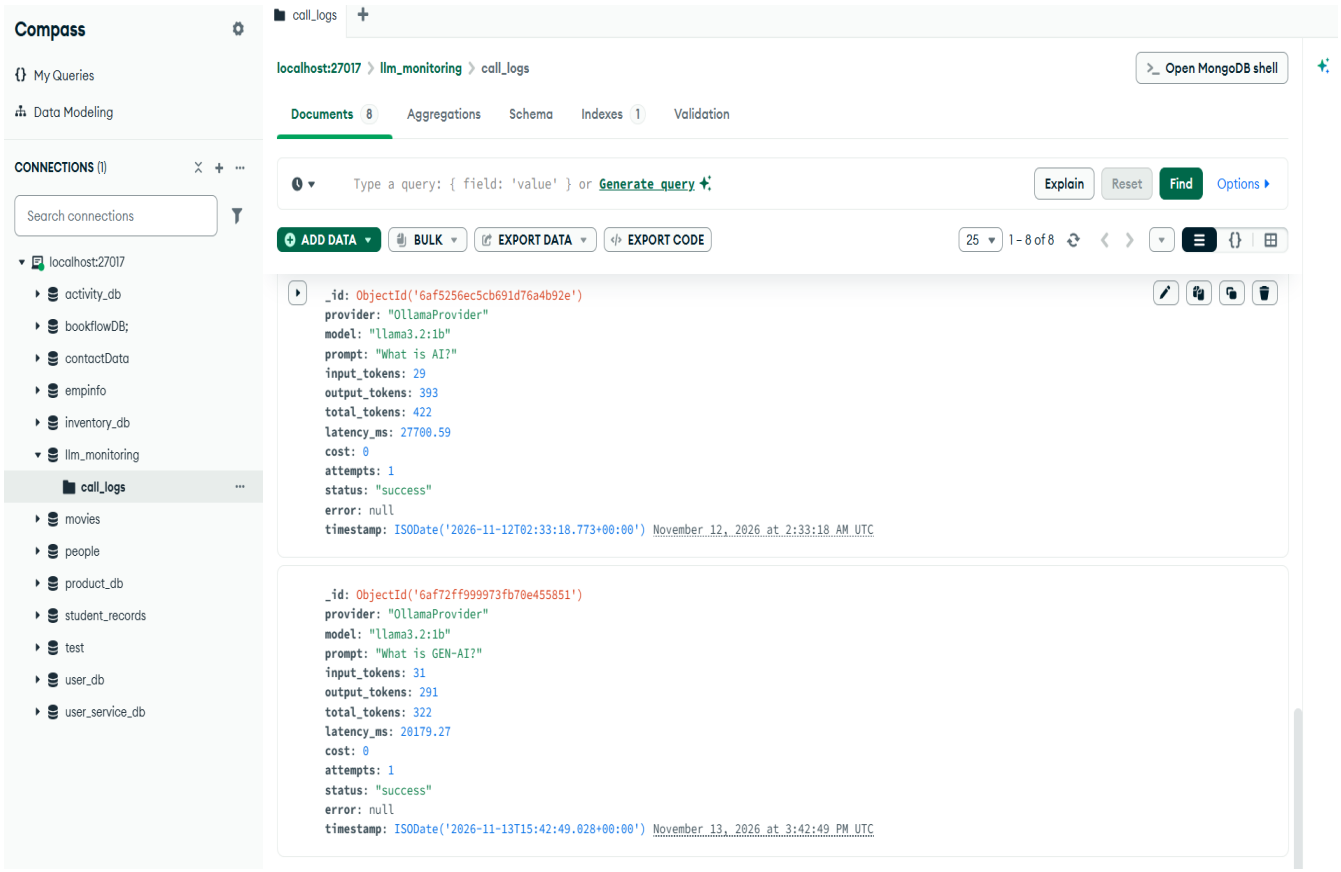


Step 7:-



OUTPUT:-


Week-3
Version Control
Current File
Unlock Pro
Project
main.py
llm_client.py
mongo_logger.py
cost_calculator.py
requirements.txt
.env
Run
main
C:\Users\Priyadarshini\PycharmProjects\Week-3\.venv\Scripts\python.exe C:\Users\Priyadarshini\PycharmProjects\Week-3\main.py
==========
WEEK 3: LLM MONITORING
==========
1. Gemini
2. Ollama
Select provider (1/2): 2
Enter your question: What is GEN-AI?
Call details saved in MongoDB!
========== AI RESPONSE ==========
GEN-AI refers to General Artificial Intelligence, which is a broad term that encompasses various approaches to developing artificial intelligence systems that can learn, reaso
The term GEN-AI is often associated with the development of more advanced AI architectures, such as:
1. Explainable AI (XAI): AI systems that can provide insights into their decision-making processes and explain their outcomes.
2. Transfer learning: AI systems that can leverage knowledge and models trained on one task to perform other related tasks.
3. Meta-learning: AI systems that can learn to learn from multiple tasks and adapt to new situations.
4. Human-AI collaboration: AI systems that can work alongside humans to improve decision-making and problem-solving.
GEN-AI is also related to the concept of "General" AI, which implies that these advanced AI systems should be able to perform a wide range of tasks, including but not limited
* Reasoning and problem-solving
* Learning and adaptation
* Perception and understanding of the world
* Natural language processing
* And more


Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
activity_db
bookflowDB;
contactData
empinfo
inventory_db
llm_monitoring
call_logs
movies
people
product_db
student_records
test
user_db
user_service_db
call_logs
localhost:27017 › llm_monitoring › call_logs
Open MongoDB shell
Documents 8
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
BULK
EXPORT DATA
EXPORT CODE
25
1 - 8 of 8
_id: ObjectId('6af5256ec5cb691d76a4b92e')
provider: "OllamaProvider"
model: "llama3.2:1b"
prompt: "What is AI?"
input_tokens: 29
output_tokens: 393
total_tokens: 422
latency_ms: 27700.59
cost: 0
attempts: 1
status: "success"
error: null
timestamp: ISODate('2026-11-12T02:33:18.773+00:00') November 12, 2026 at 2:33:18 AM UTC
_id: ObjectId('6af72ff999973fb70e455851')
provider: "OllamaProvider"
model: "llama3.2:1b"
prompt: "What is GEN-AI?"
input_tokens: 31
output_tokens: 291
total_tokens: 322
latency_ms: 20179.27
cost: 0
attempts: 1
status: "success"
error: null
timestamp: ISODate('2026-11-13T15:42:49.028+00:00') November 13, 2026 at 3:42:49 PM UTC